TT U-Net: Temporal Transformer U-Net for Motion Artifact Reduction Using PAD (Pseudo All-Phase Clinical-Dataset) in Cardiac CT

Ziheng Deng, *Member, IEEE*, Weikang Zhang, Kaile Chen, Yufu Zhou, Jiao Tian, Guotao Quan, and Jun Zhao, *Member, IEEE*

***Abstract*—Involuntary motion of the heart remains a challenge for cardiac computed tomography (CT) imaging. Although the electrocardiogram (ECG) gating strategy is widely adopted to perform CT scans at the quasi-quiescent cardiac phase, motion-induced artifacts are still unavoidable for patients with high heart rates or irregular rhythms. Dynamic cardiac CT, which provides functional information of the heart, suffers even more severe motion artifacts. In this paper, we develop a deep learning based framework for motion artifact reduction in dynamic cardiac CT. First, we build a PAD (Pseudo All-phase clinical-Dataset) based on a whole-heart motion model and single-phase cardiac CT images. This dataset provides dynamic CT images with realistic-looking motion artifacts that help to develop data-driven approaches. Second, we formulate the problem of motion artifact reduction as a video deblurring task according to its dynamic nature. A novel TT U-Net (Temporal Transformer U-Net) is proposed to excavate the spatiotemporal features for better motion artifact reduction. The self-attention mechanism along the temporal dimension effectively encodes motion information and thus aids image recovery. Experiments show that the TT U-Net trained on the proposed PAD performs well on clinical CT scans, which substantiates the effectiveness and fine generalization ability of our method. The source code, trained models, and dynamic demo will be available at https://github.com/ivy9092111111/TT-U-Net.**

***Index Terms*—Cardiac computed tomography, motion artifact reduction, deep learning, transformer.**

Manuscript received 26 July 2023; accepted 26 August 2023. Date of publication 31 August 2023; date of current version 30 November 2023. This work was supported in part by the Major Research Plan of the National Natural Science Foundation of China under Grant 92059206, in part by the Shanghai Hospital Development Center Clinical Science and Technology Innovation Project under Grant SHDC12019X22, and in part by the National Key Research and Development Program of China under Grant 2016YFC0104608. *(Corresponding author: Jun Zhao.)*

Ziheng Deng, Kaile Chen, Yufu Zhou, and Jun Zhao are with the School of Biomedical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: ZihengDeng@sjtu.edu.cn; kaile_chen@sjtu.edu.cn; henryzhouyufu@sjtu.edu.cn; junzhao@sjtu.edu.cn).

Weikang Zhang is with the School of Biomedical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China, and also with Shanghai United Imaging Healthcare Company Ltd., Shanghai 201815, China (e-mail: zhangwk9265@sjtu.edu.cn).

Jiao Tian and Guotao Quan are with Shanghai United Imaging Healthcare Company Ltd., Shanghai 201815, China (e-mail: jiao.tian@united-imaging.com; guotao.quan@united-imaging.com).

Digital Object Identifier 10.1109/TMI.2023.3310933

I. Introduction

Cardiac CT is a promising imaging technique. With the characteristics of noninvasive imaging and high spatial resolution, cardiac CT presents fine structures of hearts and plays an important role in the assessment of cardiac diseases. However, the temporal resolution of the current cardiac CT system is still limited. The resultant motion artifacts not only impede the accurate reconstruction of anatomical structures, but also block the way to functional cardiac CT [1].

Recent developments in CT hardware make it easier to perform cardiac CT and provide images with fewer motion artifacts. The most advanced multirow CT system is equipped with large-area detectors to image the whole heart within one beat [2], [3]. Fast gantry rotation is also achieved to directly improve the temporal resolution. Unfortunately, even faster gantry speeds are beyond today's mechanical limits [1]. Dual-source and multi-source scanners have been proposed for shorter acquisition times [4], [5], but with considerable cost increases. Electrocardiogram (ECG) gated scan protocols synchronize data acquisition with ECG signals, and thus CT imaging can be performed at an appropriate cardiac phase to reduce both motion artifacts and radiation dose [6], [7]. The above techniques together improve the temporal resolution of the cardiac CT system by a large margin. However, it is becoming more difficult to further improve temporal resolution with hardware alone.

The image reconstruction algorithm is an essential component of modern CT systems. For cardiac CT imaging, the motion-compensated reconstruction (MCR) method is the main solution to motion artifact reduction. With an estimated motion model, projection data collected within a wider time window can be used to reconstruct the moving object. The estimated motion model can be acquired with image-based algorithms [8], [9], partial angular reconstruction (PAR) based algorithms [10], [11], and forward projection matching methods [12], [13]. However, the difficulty in motion model optimization and high computation cost are the limitations.

Compared with traditional CT reconstruction algorithms, deep learning based methods exhibit competitive performance and are computationally efficient [14]. In 2019, a feasibility study demonstrated that CNN can achieve high accuracy for motion artifact recognition and quantification [15].

Subsequently, two different strategies were proposed for motion artifact reduction in cardiac CT. One attempted to recover the degraded images in a post-processing way [16], [17], and another followed the traditional way to model cardiac motion with deep neural networks [18], [19]. However, there are two common challenges in developing deep learning based methods for cardiac CT imaging: 1) the lack of a proper dataset to train and evaluate a data-driven model and 2) the lack of a high-capacity network model to process spatiotemporal data.

A paired dataset, which provides matched reference images and artifact-perturbed images, is the ideal dataset for deep learning research. Unfortunately, paired clinical data are extremely scarce for cardiac CT imaging. A preliminary attempt was recently made by Gong et al. [20]. They acquired forty cardiac CT exams from clinical dual-source CT systems and reconstructed images from both dual-source data and single-source data to achieve different temporal resolutions. The two sets of images are regarded as paired samples. However, the availability of dual-source CT is still limited, and there is no public dataset. Computer simulation of cardiac motion artifacts is also difficult because it relies on motion-corrected images that are both **dynamic** and **realistic**. The well-known 4D extended cardiac-torso (XCAT) phantom [21] provides a parameterized model of cardiac motion, but the appearances of the digital phantom are very different from real CT images. On the other hand, single-phase clinical cardiac CT images (usually at a quasi-quiescent cardiac phase) with real appearances are available. The researchers then used synthetic motion fields to warp the coronary artery and simulate motion artifacts [15], [18], [19]. The coronary motion artifacts were controllable, but they were confined to a relatively small region around the vessel and the motion of the myocardium was ignored.

Meanwhile, deep learning models for dynamic cardiac CT are currently underresearched. Cardiac CT potentially provides functional information about the heart (*e.g.*, dynamic myocardial perfusion CT [1] and pre-procedural dynamic CT of transcatheter mitral valve replacement [22]). Also, previous research has demonstrated that the appearance of motion artifacts manifests in a typical pattern along the temporal dimension [23]. It is reasonable to formulate motion artifact reduction as a video deblurring task and employ a spatiotemporal model. In the broader context of dynamic medical imaging, researchers have proposed several modified models to handle high-dimensional spatiotemporal information. Zhi et al. proposed a CycN-Net based on 2D+time convolution for dynamic denoising of CBCT [24]. Similarly, a 2D+time U-Net was proposed by Hauptmann [25] for cardiac MR imaging, which does not distinguish between spatial and temporal dimensions. Instead, Lyu et al. proposed an advanced bi-directional convolutional long short-term memory neural network (ConvLSTM NN) to independently model temporal information [26]. In general, a powerful network model should be compatible with the physiological characteristics of cardiac motion. Because of the nonuniform cardiac motion, images at quasi-quiescent phases yield fewer motion artifacts and should contribute more to motion artifact reduction of the whole series. Also, cardiac motion is known to be quasi-periodic, and two distant frames may share similar spatial features. Thus, the local and global temporal information are complementary. As CNN and RNN are less efficient in modeling long-range dependencies, we argue that they are suboptimal choices for cardiac CT imaging.

To address the above two problems, we propose a Pseudo All-phase clinical-Dataset (PAD) and then leverage it to train a specially designed temporal transformer U-Net (TT U-Net). Specifically, the PAD is built with a 4D cardiac motion model and various single-phase clinical CT images. Thus, it not only contains the motion of the **whole heart** for **a complete cardiac cycle** but also exhibits **realistic appearances**. In addition, the PAD provides paired samples for **supervised learning and quantitative evaluation** of a data-driven model. Based on the PAD, we further design and train a novel TT U-Net for motion artifact reduction in cardiac CT. Within the TT U-Net, the temporal transformer layer (TTL) is proposed to integrate the **global and local temporal features** with self-attention. Comprehensive experiments on the PAD and real clinical cardiac CT scans demonstrate the effectiveness and generalization ability of our method.

II. Method

In this section, we first formulate the motion artifact reduction problem in cardiac CT. After that, the main components of the proposed framework including the PAD and the TT U-Net model are described.

A. Motion Artifact Reduction Problem Formulation

Cardiac CT image reconstruction is regarded as a dynamic inverse problem because the object being imaged (the heart) undergoes a temporal evolution during data acquisition. Specifically, the mathematical expression of cardiac CT imaging is given by:

$$\boldsymbol{p}(t) = \boldsymbol{A}(t)\boldsymbol{f}(t) + \boldsymbol{e}(t), \tag{1}$$

where $\boldsymbol{p}(t)$ are the acquired projection data at time t, $\boldsymbol{A}(t)$ is the forward projection operator, $\boldsymbol{f}(t)$ is the dynamic image, and $\boldsymbol{e}(t)$ is the measurement error. A reconstructed image based on the short-scan Feldkamp-Davis-Kress (FDK) algorithm [27] is given by:

$$\boldsymbol{f}_{FDK}(t_s) = \sum_{t \in T_s} \boldsymbol{A}^{-1}(t)\boldsymbol{W}(t, \boldsymbol{p}'(t)), \tag{2}$$

where t_s is the selected reconstruction phase, T_s is the time window centered around t_s and in which enough projection data are acquired for short-scan reconstruction, $\boldsymbol{A}^{-1}(t)$ is the backporjection operator, $\boldsymbol{p}'(t)$ refers to the filtered projection data, and $\boldsymbol{W}(\cdot)$ is the weighting function for short-scan FDK reconstruction. Motion artifacts are observed on $\boldsymbol{f}_{FDK}$ due to spatiotemporal ambiguities.

To implement MCR methods, the dynamic image $\boldsymbol{f}(t)$ is reformulated as a time-varying deformation of the target image $\boldsymbol{f}(t_s)$. The deformation field $\boldsymbol{DF}(t_s, t)$ describes the cardiac motion from the selected phase t_s to arbitrary phase t:

$$\boldsymbol{f}(t) = \boldsymbol{DF}(t_s, t) \circ \boldsymbol{f}(t_s), \tag{3}$$

where $\circ$ is the image warping operation. If an estimated motion model $\tilde{\boldsymbol{DF}}(t_s, t)$ exists, MCR is achieved by slightly modifying (2). For example, the backprojection-then-warp (BPW) method [28] helps reduce motion artifacts as follows:

$$\boldsymbol{f}_{BPW}(t_s) = \sum_{t \in T_s} \tilde{\boldsymbol{DF}}^{-1}(t_s, t) \circ (\boldsymbol{A}^{-1}(t)\boldsymbol{W}(t, \boldsymbol{p}'(t))). \quad (4)$$

Meanwhile, recent deep learning based methods also allow motion artifact reduction in the absence of a motion model. In this context, it is usually formulated as an image post-processing problem. In general, the learning based problem can be modeled as follows:

$$arg \min_{\boldsymbol{\theta}} \frac{1}{N_\Omega} \sum_{\{\hat{f}f\} \in \Omega} \mathcal{L}(\boldsymbol{G}(\hat{\boldsymbol{f}}, \boldsymbol{\theta}), \boldsymbol{f}), \quad (5)$$

where $\boldsymbol{G}(\cdot, \boldsymbol{\theta})$ refers to an image deblurring model with trainable parameters $\boldsymbol{\theta}$, $\hat{\boldsymbol{f}}$ is the degraded image (e.g., $\boldsymbol{f}_{FDK}$), and $\mathcal{L}(\mathbf{G}(\hat{\boldsymbol{f}}, \boldsymbol{\theta}), \boldsymbol{f})$ measures the discrepancy between the restored image and the ground truth image $\boldsymbol{f}$. The learned parameters $\boldsymbol{\theta}$ are obtained through the optimization of (5) on a proper dataset Ω that contains N_Ω training pairs.

We will show how to establish a dataset Ω and then train a tailored network model for motion artifact reduction in the following subsections.

B. Pseudo All-Phase Clinical-Dataset (PAD)

The PAD provides paired pseudo all-phase clinical cardiac CT images. Cardiac motion artifacts originate from the relative motion between the heart and the CT system. Thus, dynamic cardiac CT images with few motion artifacts become the prerequisite for motion artifact simulation. Due to the scarcity of such real data, we begin by synthesizing reference pseudo all-phase clinical CT images. We expect the synthetic images to be both dynamic and realistic. Intuitively, it is achieved by animating a single-phase clinical image with an all-phase cardiac motion model (see the workflow in Fig. 1). The artifact-perturbed pseudo all-phase clinical CT images are then generated by computer-simulated CT scan according to (1), and (2). In this paper, the cardiac motion model is extracted from the XCAT phantom [21], and the realistic cardiac anatomy comes from the MICCAI 2017 Multi-Modality Whole Heart Segmentation (MM-WHS) dataset [29]. The PAD can be easily extended by involving more single-phase clinical CT images.

1) Cardiac Motion Model: An artificial motion field is needed to animate the single-phase clinical CT (Fig. 1B). We build a 4D statistical shape model (4D SSM) to generate personalized cardiac motion. The 4D SSM is able to describe the spatiotemporal distribution of anatomically corresponding landmarks from a set of training shapes. To build the 4D SSM, we involve 21 4D XCAT phantoms corresponding to 21 volunteers. Empirically, for each 4D phantom, 3D volumes are sampled at 5% R-R intervals throughout the cardiac cycle, resulting in 20 volumes to depict a whole cardiac cycle. We first convert each 3D volume to a point distribution model (PDM) and align them in a reference coordinate system. The 4D SSM is built following Perperidis's method [30]. If a clinical 4D dataset is available, one may follow Hoogendoorn's work [31] to build the 4D SSM.

Let $\left\{\boldsymbol{s}^{i,j}_{XCAT} \in \mathbb{R}^{3M} | i = 1, \ldots, N_i; j = 1, \ldots, N_j\right\}$ denote the shapes of N_i 4D XCAT phantoms each with N_j phases. Each shape $\boldsymbol{s}^{i,j}_{XCAT}$ in the $3M$-dimensional space consists of M 3D surface landmarks. Principal component analysis (PCA) is adopted to describe the various anatomical structures and cardiac motions among these samples:

$$\boldsymbol{s}^{i,j}_{XCAT}(\boldsymbol{b}^i_{anatomy}, \boldsymbol{b}^{i,j}_{motion}) = \bar{\boldsymbol{s}} + \boldsymbol{P}_{anatomy}\boldsymbol{b}^i_{anatomy} + \boldsymbol{P}_{motion}\boldsymbol{b}^{i,j}_{motion}, \quad (6)$$

where $\bar{\boldsymbol{s}}$ is the mean shape. $\boldsymbol{P}_{anatomy} \in \mathbb{R}^{3M \times N_a}$ describes the **cardiac anatomy variability** among different volunteers. The columns of $\boldsymbol{P}_{anatomy}$ are orthonormal eigenvectors of the covariance matrix:

$$\boldsymbol{C}_{anatomy} = \frac{1}{N_i} \sum_{i=1}^{N_i} (\boldsymbol{s}^{i,*} - \bar{\boldsymbol{s}})(\boldsymbol{s}^{i,*} - \bar{\boldsymbol{s}})^T, \quad (7)$$

where $\boldsymbol{s}^{i,*}$ is the shape at the reference phase for each case.

The $\boldsymbol{b}^i_{anatomy}$ in (6) are weights for different anatomy modes and are obtained by projecting the normalized shapes $\boldsymbol{S}^i = (\boldsymbol{s}^{i,1} - \bar{\boldsymbol{s}}, \boldsymbol{s}^{i,2} - \bar{\boldsymbol{s}}, \ldots, \boldsymbol{s}^{i,N_j} - \bar{\boldsymbol{s}})$ onto the eigenmodes $\boldsymbol{P}_{anatomy}$. The $\boldsymbol{P}_{motion} \in \mathbb{R}^{3M \times N_m}$ in (6) describes the **cardiac motion variability** among different cardiac phases. The columns of $\boldsymbol{P}_{motion}$ are orthonormal eigenvectors of the covariance matrix:

$$\boldsymbol{C}_{motion} = \frac{1}{N_i N_j} \sum_{i=1}^{N_i} \sum_{j=1}^{N_j} (\boldsymbol{s}^{i,j} - \boldsymbol{s}^{i,*})(\boldsymbol{s}^{i,j} - \boldsymbol{s}^{i,*})^T. \quad (8)$$

To estimate the motion weights $\boldsymbol{b}^{i,j}_{motion}$ in (6), we project the displacement vectors $\boldsymbol{S}_{dis} = (\boldsymbol{s}^{i,1} - \boldsymbol{s}^{i,*}, \boldsymbol{s}^{i,2} - \boldsymbol{s}^{i,*}, \ldots, \boldsymbol{s}^{i,N_j} - \boldsymbol{s}^{i,*})$ within a cardiac cycle onto the eigenmodes $\boldsymbol{P}_{motion}$.

One way to utilize the motion information in (6) and make a single-phase clinical CT image to "beat again" is to adopt the average motion among training XCAT samples:

$$\boldsymbol{s}^j_{PAD} = \boldsymbol{s}_{clinical} + \boldsymbol{P}_{motion}\overline{\boldsymbol{b}^j_{motion}}, \quad (9)$$

where $\boldsymbol{s}^j_{PAD}$ is the predicted shape model at j^{th} phase, $\boldsymbol{s}_{clinical}$ is the normalized shape model of the clinical image at its original phase, and $\overline{\boldsymbol{b}^j_{motion}}$ is the average motion weight at j^{th} phase. An alternative way is to further perform PCA on the motion weights [32] to generate various motion models. Thus far, we have obtained a **dynamic shape model** from a single-phase clinical image (Fig. 1B). In previous studies, the dynamic shape model is then used for the evaluation of heart function. In the research of CT image reconstruction, we take one step further to synthesize **pseudo all-phase CT images**.

2) Synthesizing Pseudo All-Phase Cardiac CT From Single-Phase Clinical Images: We use a deep learning based ShapeMorph model to fill in the gap between the dynamic shape model and pseudo all-phase CT images.

The dynamic shape model is converted to image volumes in the form of segmentation maps and input to the ShapeMorph model (Fig. 1C). The ShapeMorph is a modified VoxelMorph model [33]. Instead of taking a pair of 3D images as input, it predicts a dense deformation field from a pair of segmentation maps. The ShapeMorph model is trained on the XCAT dataset in a self-supervised manner similar to VoxelMorph. A sequence of deformation fields $\boldsymbol{DF}_{PAD}(t_{clinical}, t)$ is obtained by registering each segmentation map with the one at the reference phase (Fig. 1D). The 4D deformation field is

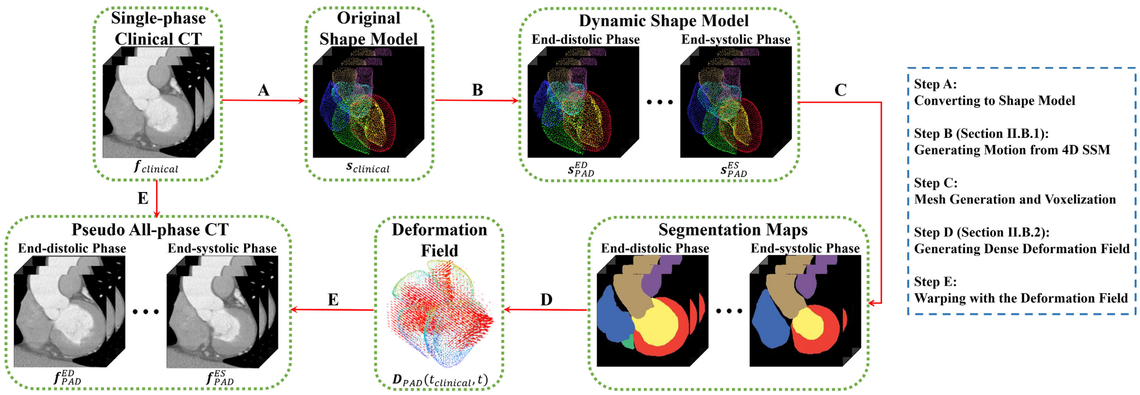

Fig. 1. Flowchart of synthesizing pseudo all-phase cardiac CT from single-phase images.

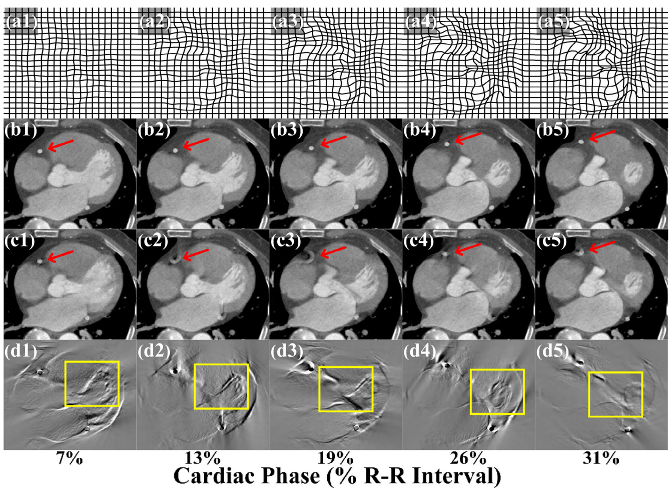

Fig. 2. Simulation process of the PAD. (a1)-(a5) Pseudo all-phase deformation fields, (b1)-(b5) reference pseudo all-phase clinical images, (c1)-(c5) artifact-perturbed pseudo all-phase clinical images, (d1)-(d5) difference images that show the whole-heart motion artifacts. The coronary arteries are denoted by red arrows. The myocardium motion artifacts are labeled by yellow boxes.

interpolated for higher temporal resolution and then used to warp the original single-phase clinical CT image $\boldsymbol{f}_{clinical}$ to produce pseudo all-phase CT images (Fig. 1E):

$$\boldsymbol{f}^{t}_{PAD} = \boldsymbol{DF}_{PAD}(t_{clinical}, t) \circ \boldsymbol{f}_{clinical}. \tag{10}$$

3) Cardiac Motion Artifact Simulation: Cardiac motion artifact simulation: Example images of the PAD are shown in Fig. 2. The pseudo all-phase deformation fields (Fig. 2(a1)-(a5)) describe the cardiac motion of the reference pseudo all-phase images (Fig. 2(b1)-(b5)). With the reference pseudo all-phase images, we then acquire artifact-perturbed images using a computer-simulated CT scan. Specifically, cone-beam projection data are simulated using (1) with a realistic CT scan geometry. Because of the nonnegligible time of gantry rotation, the projection data are collected from the reference images at different phases. Short-scan FDK reconstruction is performed using (2). As image reconstruction requires projection data collected within a time window T_s (in most cases, T_s is no shorter than half a gantry rotation period), the reconstructed image will contain mixed spatiotemporal information and exhibit a blurry appearance. According to the CT scan geometry, the artifact-perturbed images will further include irregular coronary motion artifacts such as crescent (Fig. 2(c2)-(c3)), tail (Fig. 2(c5)), and horn (Fig. 2(c4)) artifacts [23]. Images with realistic whole-heart motion artifacts and the corresponding ground truth are achieved with our method. In Fig. 3, we present a visual comparison between the simulated PAD images and the clinical multiphase cardiac CT images. It can be observed that the PAD provides high-quality images, and the synthetic and clinical images are visually indistinguishable. The PAD images exhibit realistic motion artifacts, and even the evolution patterns of the coronary artifacts are quite similar.

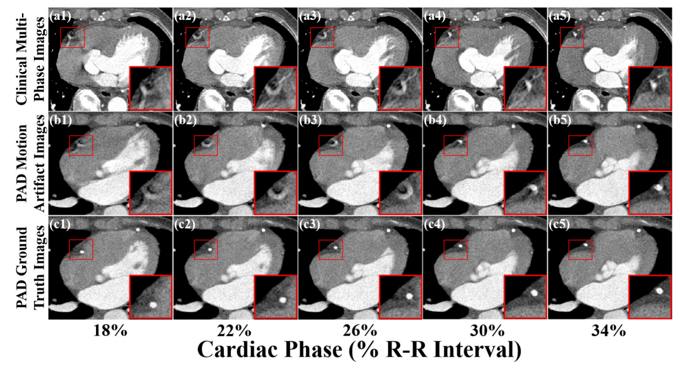

Fig. 3. A visual comparison between the PAD and clinical multiphase cardiac CT images. (a1)-(a5) Clinical multiphase cardiac CT images, (b1)-(b5) PAD motion artifact images, (c1)-(c5) PAD ground truth images.

C. TT U-Net

The dynamic nature of cardiac CT distinguishes it from other CT imaging problems. With the help of the PAD, it is possible to model and utilize the spatiotemporal correlation in cardiac CT images in a data-driven manner. Thus, we propose a novel TT U-Net for dynamic data. First, we extend (5) to a video deblurring problem:

$$arg \min_{\boldsymbol{\theta}} \frac{1}{N_\Omega} \sum_{\left\{\hat{\boldsymbol{F}}, \boldsymbol{F}\right\} \in \Omega} \mathcal{L}(\boldsymbol{G}(\hat{\boldsymbol{F}}, \boldsymbol{\theta}), \boldsymbol{F}), \tag{11}$$

where $\hat{\boldsymbol{F}} = \left\{\hat{\boldsymbol{f}}(t_1), \hat{\boldsymbol{f}}(t_2), \ldots, \hat{\boldsymbol{f}}(t_{N_c})\right\}$ represents a video clip of degraded cardiac CT and $\boldsymbol{F} = \left\{\boldsymbol{f}(t_1), \boldsymbol{f}(t_2), \ldots, \boldsymbol{f}(t_{N_c})\right\}$ represents the ground truth video clip. N_c is the length of each video clip. In this paper, N_c is set to 48.

1) Overall Architecture: The architecture of TT U-Net is shown in Fig. 4. U-Net [34] has proven to be an effective and

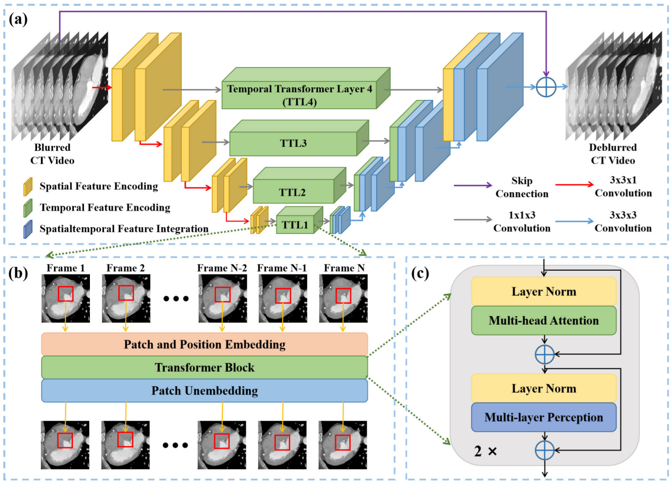

Fig. 4. The architecture of the proposed TT U-Net. (a) The TT U-Net with temporal transformer layers (TTL), (b) The data flow of the TTL, (c) The architecture of the transformer block inside the TTL.

lightweight model for image deblurring and hence it serves as the backbone of the proposed model. The hierarchical architecture is suitably flexible to model spatial information and provides multi-resolution feature maps. To capture the temporal relation among the dynamic CT images, we replace the skip connection module with the proposed temporal transformer layer (TTL), which introduces a self-attention mechanism to temporal modeling. As the TTL operates mainly in the temporal dimension, the model achieves better performance with high efficiency. Note that we omit the TTL in the layer of the highest resolution in the experiments to achieve faster training and inference (please refer to Section IV-C for more information). For the decoder part of the U-shape neural network, 3D convolution is used to integrate the spatiotemporal features from each resolution.

2) Temporal Transformer Layer (TTL): The proposed TTL contains a patch and position embedding layer, a standard transformer encoder block, and a patch unembedding layer, as shown in Fig. 4(b). Different from the famous ViT [35] and Swin Transformer [36], the proposed TTL focuses on temporal modeling. To this end, instead of computing self-attention within the whole image or a spatial window, TTL conducts self-attention in a temporal window. The input of the TTL is a spatiotemporal feature map $\mathcal{X} \in \mathbb{R}^{B\times N_c\times C\times H\times W}$ where B, N_c, C, H, W denotes the batch size, number of phases, number of channels, height, and width of the feature map, respectively. Patch and position embedding layers are used to convert the high-dimensional feature map to a sequence of patch embeddings. Specifically, we first split the feature map into nonoverlapping $\frac{BHW}{P^2} \times N_c \times C \times P \times P$ patches in the spatial dimension. After that, each high-dimensional patch is flattened except for the temporal dimension and converted to an $N_C \times CPP$ embedding. Each token is further projected to an $N_C \times D$ space with a trainable projection layer. The patch embedding layer helps enlarge the spatial receptive field with low computational complexity. To retain the positional information, we also add position embeddings to the patch embeddings. Two cascaded transformer blocks thereafter integrate and process the spatiotemporal information. Within the transformer block, standard multihead self-attention is conducted in a temporal window of size N_C to explore the global interactions among different cardiac phases. The patch unembedding layer performs a reshape operation. It recovers the feature map from the processed patch embeddings and provides the input to the decoder part of the U-Net.

III. Experiments

A. Comparison Methods

Five well-known algorithms are involved for comparison. 1) Parker weighted short scan FDK method [27]. 2) PAR method [10], which is an efficient MCR method with high performance. 3) 2D+time U-Net [25], which is a fully convolutional neural network designed for dynamic imaging. 4) ConvLSTM NN [26], which is an LSTM-based neural network designed for dynamic imaging. 5) VRT [37], which is a transformer-based neural network designed for natural video restoration. The VRT uses the temporal mutual self-attention (TMSA) module to simultaneously encode spatial and temporal information. As the temporal window size is limited for each TMSA module, the VRT adopts the shift window strategy to progressively enlarge the temporal receptive field. By comparison, the TT U-Net can efficiently model long-range dependencies within each TTL. The last three data-driven methods require a training process before application.

B. Datasets

1) PAD: To establish the PAD, we selected 43 cases of single-phase clinical cardiac CT images from the MM-WHS dataset [29]. As some of these images lacked segmentation annotations, we first segmented these images manually. We then used the 4D SSM to convert the single-phase images into pseudo all-phase clinical images as described in Section II-B. For each single-phase CT image, we generated three sets of reference pseudo all-phase images with different heart rates. Then, we simulated axial cone-beam CT scans to obtain artifact-perturbed CT images. The geometry configuration was as follows. The distances from the source and detector to the rotation center were both 1000 mm. The detector had a physical size of 600 mm$\times$150 mm with 800 $\times$ 200 channels. The view sampling rate was 1000 projection views per turn. The gantry rotation speed was 250 ms per turn, which led to a temporal resolution of around 135 ms if short-scan reconstruction is adopted. For each case, the simulated scan started at a random cardiac phase and random gantry angle to simulate an ECG-free scan protocol and provide various motion artifacts. The reconstructed image had a size of 512 $\times$ 512$\times$120 voxels. We cropped and resized the region of interest (ROI) to a volume of 256 $\times$ 256$\times$90 voxels. 48 frames of images with a time interval of 12.5 ms were reconstructed for dynamic CT imaging. The images were regrouped slice by slice to form video clips each with 48 frames of 256 $\times$ 256 pixel images.

The PAD was randomly separated into 3 groups at the patient level: 37 for the training set, 3 for the validation set, and 3 for the test set. The training set and the validation set were used to train the aforementioned neural networks. The testing set was used to quantitatively evaluate the performance of all the methods.

TABLE I
REAL CLINICAL DATASET USED IN THE EXPERIMENT

Clinical Case	Heart Rate (bpm)	Phase Range (R-R Interval)	Reconstruction Frames	Reconstruction Time Interval (ms)
Case 1	79	30% ∼ 90%	48	10.0
Case 2	71	30% ∼ 90%	48	10.4
Case 3	71	24% ∼ 90%	48	11.7
Case 4	77	15% ∼ 60%	48	7.5
Case 5	83	30% ∼ 50%	20	7.2

Abbreviation: bpm - beats per minute.

2) Real Clinical Dataset: Five clinical cardiac CT scans were retrospectively collected. Projection data were acquired with a 320-row CT scanner (uCT 960+, United Imaging Healthcare) following an ECG-gated axial scan. The gantry rotation speed was 250 ms per turn. The scans generally started at the mid-systole phase and ended at the end-diastole phase but varied from case to case. To implement the proposed method and other deep learning based comparison methods, we first reconstructed the images using the Parker weighted short scan FDK method. Proper reconstruction intervals are set to acquire an evenly spaced image series of 48 frames except for case 5. Then, we cropped the heart ROI and reformatted the uncorrected images in a slice-by-slice manner to serve as the input of the network model. More detailed information is listed in Table I. The real clinical dataset was used to evaluate the performance of all methods. Additionally, it was used to evaluate the generalization ability of the TT U-Net trained on the PAD.

C. Implementation Details

For a fair comparison, we used the same dataset and loss functions to train all the neural networks. Both the pixelwise $L1$ loss and the WGAN-GP adversarial loss were used. The $L1$ loss worked well for recovering low-frequency signals, whereas the adversarial loss helped to better reconstruct high-frequency details [38]. We used a Markovian discriminator [39] $\boldsymbol{D}(\cdot, \boldsymbol{\theta_D})$ with trainable parameters $\boldsymbol{\theta_D}$ for the WGAN training process. The overall objective function for the deblurring model $\boldsymbol{G}(\cdot, \boldsymbol{\theta})$ was the combination of the $L1$ loss and the adversarial loss:

$$arg \min_{\boldsymbol{\theta}} \frac{1}{N_\Omega} \sum_{\left\{\hat{\boldsymbol{F}},\boldsymbol{F}\right\} \in \Omega} [\mathcal{L}1(\boldsymbol{G}(\hat{\boldsymbol{F}}, \boldsymbol{\theta}), \boldsymbol{F}) - \lambda_1 \mathbb{E}(\boldsymbol{D}(\boldsymbol{G}(\hat{\boldsymbol{F}}, \boldsymbol{\theta}), \boldsymbol{\theta_D})], \tag{12}$$

where λ_1 was empirically set to 0.001. The objective function for the discriminator is the combination of the Wasserstein distance (the first two terms) and the gradient penalty (the third term) [26]:

$$arg \min_{\boldsymbol{\theta_D}} \frac{1}{N_\Omega} \sum_{\left\{\hat{\boldsymbol{F}},\boldsymbol{F}\right\} \in \Omega} [\mathbb{E}(\boldsymbol{D}(\boldsymbol{G}(\hat{\boldsymbol{F}}, \boldsymbol{\theta}), \boldsymbol{\theta_D})) - \mathbb{E}(\boldsymbol{D}(\boldsymbol{F}, \boldsymbol{\theta_D})) + \lambda_2 \mathbb{E}_{\widetilde{\boldsymbol{F}}} \parallel \nabla_{\widetilde{\boldsymbol{F}}} \boldsymbol{D}(\widetilde{\boldsymbol{F}}, \boldsymbol{\theta_D}) - 1 \parallel_2^2], \tag{13}$$

where ∇ is the gradient operator. $\widetilde{\boldsymbol{F}}$ is uniformly sampled along straight lines connecting pairs of generated samples $\boldsymbol{G}(\hat{\boldsymbol{F}}, \boldsymbol{\theta})$ and real samples $\boldsymbol{F}$, and λ_2 was empirically set to 10.

TABLE II
THE NUMBER OF PARAMETERS (# PARAMS) AND COMPUTATIONAL COST OF DIFFERENT NETWORK MODELS

Metric	2D+time U-Net	ConvLSTM NN	VRT	TT U-Net (proposed)
# Params (million)	2.10	1.70	12.30	24.88
FLOPs (G/frame)	18.75	30.80	50.46	46.02
Test time (ms/frame)	3.87	10.76	21.54	11.17

We implemented all the deep learning based models using the PyTorch toolbox, and the training was performed on an NVIDIA Tesla A100 GPU with 40 GB memory. The models were optimized using the Adam algorithm. The learning rate slowly decreased from 10^{-4} to 10^{-5}. The batch size was set to 1 because of limited memory. Note that we simplified the structure of the ConvLSTM NN and the VRT because of limited memory (the model was designed for either 7 frames of 100×100 images or 24 frames of 64×64 images in the original papers whereas 48 frames of 256×256 images are used as the input here). The number of parameters and computational cost of the neural network models are listed in Table II. The test time was measured on an NVIDIA RTX 2080 Ti GPU with 11 GB memory. The CT reconstruction was implemented using MATLAB R2021a.

D. Evaluation Metrics

The simulated PAD is a paired dataset and thus allows quantitative evaluation. The structural similarity (SSIM) index and the root mean squared error (RMSE) were adopted to evaluate various methods. As cardiac motion artifacts mainly appear in the regions of structure boundaries (for example, artifacts shown in the yellow boxes of Fig. 2), we calculated the SSIM and RMSE within these local patches. In addition, we segmented the right coronary artery (RCA) of each case and calculated the Dice score according to the ground truth image to evaluate the performance of coronary motion artifact reduction.

Although ground truth images do not exist for the real clinical dataset, several motion artifact metrics have been proposed to quantify the visual quality of the reconstructed coronary arteries. We adopted the fold overlap ratio (FOR), low-intensity region score (LIRS), and motion artifact score (MAS) metrics to evaluate the proposed method on the real clinical dataset. These metrics are shown to provide high agreement in motion artifact ranking when compared with expert readers [23]. We focused on RCAs in this experiment.

The FOR measures the shape of the coronary artery. Given a binary image of the segmented vessel, two orthogonal axes v_1 and v_2 passing through the segmented vessel centroid are determined by evaluating the principal component eigenvectors of the vessel pixels [16]. Each axial will divide the vessel into two regions $R_1^{v_i}$ and $R_2^{v_i}$. By mirroring $R_1^{v_i}$ according to the axial v_i, the overlap ratio $L_{FOR}^{v_i}$ measures the symmetrical degree of the vessel:

$$L_{FOR}^{v_i} = \parallel \hat{R_1^{v_i}} \cap R_2^{v_i} \parallel_0 / \parallel \hat{R_1^{v_i}} \cup R_2^{v_i} \parallel_0, \tag{14}$$

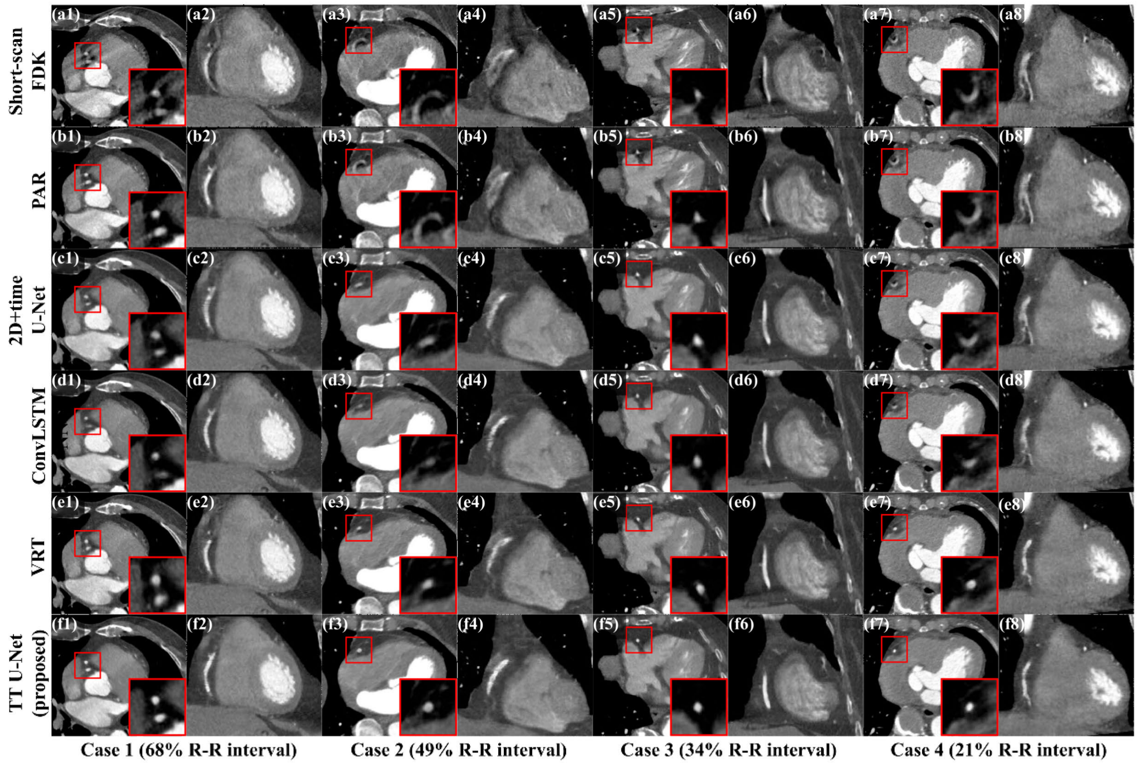

Fig. 5. Motion artifact reduction results on the clinical dataset arranged as follows: Parker weighted short-scan FDK method (a), PAR method (b), 2D+time U-Net (c), ConvLSTM NN(d), VRT (e), and TT U-Net (f). The first column of each case presents the axial images with zoom-in ROIs, and the second column presents the sagittal images. The display window is [-300,500] HU for reconstructed images, and [-100,600] HU for zoom-in ROIs.

where $\hat{R}_1^{v_i}$ is the mirrored region. The final FOR is defined as the smaller overlap ratio according to the two axes:

$$L_{FOR} = min(L_{FOR}^{v_1}, L_{FOR}^{v_2}). \qquad (15)$$

The LIRS measures the area and intensity of the motion-induced shaded region. In our implementation, we segment the low-intensity region with a threshold of -200 HU. The LIRS metric is expressed as:

$$L_{LIRS} = 0.5 \times (1 - \frac{A_{LIR}}{A_{ves}}) + 0.5 \times \frac{\overline{I_{LIR}} + 1024}{\overline{I_{BG}} + 1024}, \qquad (16)$$

where A_{LIR} is the area of the low-intensity region, A_{ves} is the area of the vessel region, $\overline{I_{LIR}}$ is the mean CT value of the low-intensity region, and $\overline{I_{BG}}$ is the mean CT value of the background myocardium region. Finally, the MAS is defined as the product of the FOR and LIRS metrics to consider the vessel shape and the motion-induced low-intensity region.

IV. Results

A. Experiments on Real Clinical Dataset

The effectiveness of the TT U-Net trained on the PAD was evaluated on 5 real clinical scans without fine-tuning. In this paper, we aim to address the challenging problem of all-phase motion artifact reduction. Here, we present reconstructions at various cardiac phases (case 1 and case 2 at the diastolic phase, case 3 and case 4 at the systolic phase), even though some of them are typically excluded from clinical usage for now. Fig. 5 presents the motion artifact reduction results of various methods in both axial and sagittal views. Severe motion artifacts are observed at rapid phases in short-scan FDK images (Fig. 5(a1)-(a8)). The PAR method estimates cardiac motion with a pair of conjugated PAR images to compensate for cardiac motion. The shape of the myocardium is partially recovered, but tail- and crescent-shaped coronary motion artifacts still exist due to rapid motion (Fig. 5(b1), (b3), (b7)). Compared with traditional reconstruction methods, deep learning approaches achieve better visual performance. Note that all the neural networks were trained on the PAD, which proves the effectiveness of the proposed dataset. The 2D+time U-Net, ConvLSTM NN, and VRT all provide images with improved quality. However, some structural details are slightly disrupted. As shown in Fig. 5(c3), (c7), the RCAs are distorted. Blurred RCAs are also observed in Fig. 5(d3), (d7), (e3). Our TT U-Net shows superior performance on both myocardium recovery and RCA reconstruction (Fig. 5(f1)-(f8)).

To quantitatively evaluate the visual quality of the reconstructed images, we present the motion artifact metrics in Table III. The results show that neural networks trained on the proposed PAD can effectively improve the visual quality of real clinical cardiac CT images. Moreover, the TT U-Net achieved the highest FOR, LIRS, and MAS metrics. In summary, the proposed TT U-Net trained on the PAD can generate satisfying results on the real clinical dataset.

Accurate reconstruction of coronary arteries is of significant clinical value. We present the straightened curved planar reformation (CPR) results of clinical cases 1-4 in Fig. 6. Shading, blurring, and doubling artifacts are visible on the CPRs of the uncorrected images, whereas the corrected ones are brighter and cleaner. Although the TT U-Net processed the images in a slice-by-slice way, the motion artifact reduction results are relatively consistent across sections.

TABLE III
AVERAGED MOTION ARTIFACT METRICS (MEANS ± SDS) OF VARIOUS METHODS ON THE CLINICAL DATASET

Metric	Short-scan FDK	PAR	2D+time U-Net	ConvLSTM NN	VRT	TT U-Net (proposed)
FOR↑	0.6304±0.0471	0.7298±0.0526	0.8033±0.0248	0.8299±0.0448	0.8266±0.0291	**0.8484±0.0221**
LIRS↑	0.5188±0.1052	0.6350±0.1259	0.7226±0.1531	0.7267±0.1967	0.7555±0.0975	**0.7834±0.1076**
MAS↑	0.3298±0.0838	0.4679±0.1203	0.5821±0.1324	0.6087±0.1886	0.6253±0.0913	**0.6660±0.1046**

Note: The ↑ means the higher, the better.

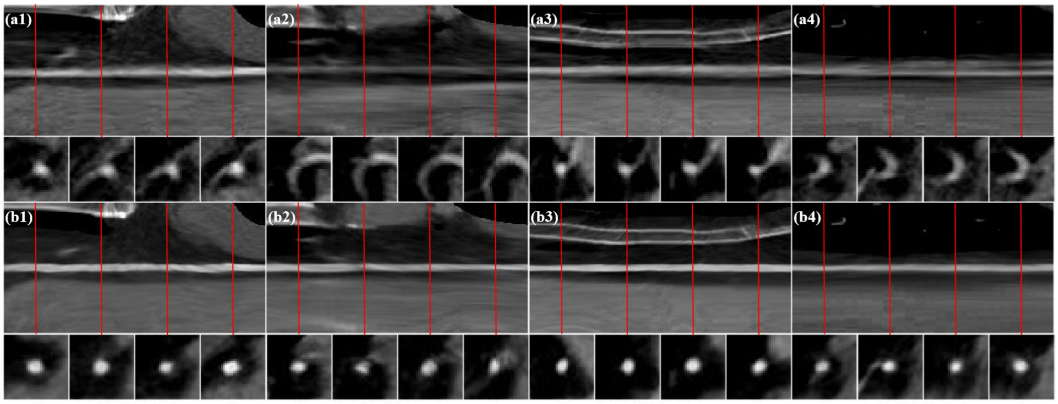

Fig. 6. Straightened curved planar reformation (CPR) results. (a1)-(a4) Results of uncorrected images, (b1)-(b4) results of TT U-Net corrected images. In each sub-image, the upper part is the straightened CPR, and the lower part indicates the cross-sectional patches corresponding to the red lines.

TABLE IV
FID OF SIMULATED MOTION ARTIFACT DATASETS

Metric	PAD Motion Artifact Images	XCAT Motion Artifact Images
FID↓	69.80	195.28

Note: For each dataset, we randomly drew 10000 2D images to calculate the FID. The ↓ means the lower, the better.

Except for the realistic cardiac motion, we argue that the effectiveness and fine generalization ability of the trained TT U-Net is also attributed to the realistic appearance of the PAD. For comparison, we built a dynamic XCAT dataset. Using XCAT software, we generated various 4D cardiac CT phantoms. Then, paired XCAT images with and without motion artifacts are simulated with the same strategy as we used for the PAD (described in Section II-B. 3). As we build the PAD with the motion model extracted from the XCAT phantom, the PAD and the XCAT dataset share a similar motion pattern. However, the images in the PAD are derived from clinical CT images and thus have a realistic appearance, while XCAT images are ideal digital phantoms with few textures. We quantify the similarity between the two simulated datasets and the real clinical dataset using the Fréchet inception distance (FID) metric, which was shown to correlate well with human judgment of visual quality [40]. FID is calculated by computing the Fréchet distance between two Gaussians fitted to feature representations of the Inception network as follows:

$$FID(\boldsymbol{x},\boldsymbol{r}) = \| \mu_{\boldsymbol{x}} - \mu_{\boldsymbol{r}} \|_2^2 + Tr(\boldsymbol{C_x} + \boldsymbol{C_r} - 2(\boldsymbol{C_x}\boldsymbol{C_r})^{1/2}), \tag{17}$$

where $\boldsymbol{x}, \boldsymbol{r}$ are feature representations of samples from the simulated datasets and the real clinical dataset, respectively.

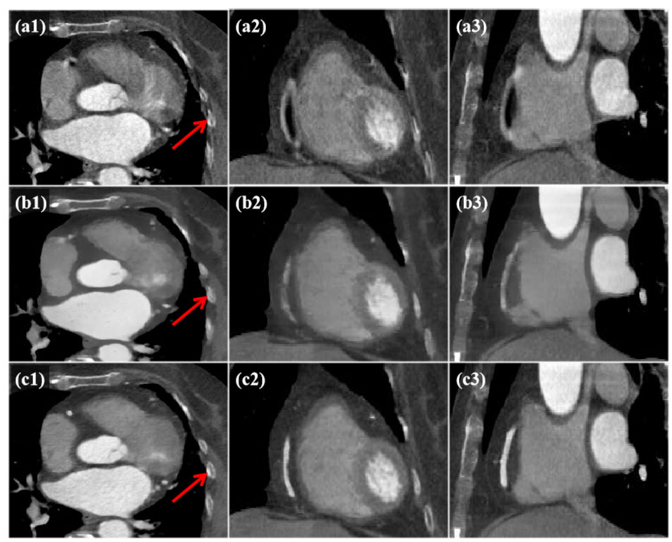

Fig. 7. Clinical case 1, motion artifact reduction results of TT U-Net trained on different datasets. (a1)-(a3) Uncorrected images, (b1)-(b3) results of model trained on the XCAT dataset, (c1)-(c3) results of model trained on the PAD.

$\mu_{\boldsymbol{x}}, \mu_{\boldsymbol{r}}$ are the means of the representations. Tr(·) calculates the trace of a matrix. $\boldsymbol{C_x}, \boldsymbol{C_r}$ are the covariance matrices of the representations. A lower FID indicates a higher similarity between the simulated dataset and the real clinical dataset. As shown in Table IV, the PAD achieves much lower FID compared with the XCAT dataset.

We further trained a TT U-Net on the dynamic XCAT dataset. The same number of training samples are included in the XCAT dataset to train the model. As shown in Fig. 7, the motion artifact reduction results of TT U-Net trained on the XCAT dataset are over-smoothed. The bone area denoted by

TABLE V
QUANTITATIVE EVALUATION (MEANS ± SDS) OF VARIOUS METHODS FOR EACH SIMULATED CASE

Simulated Case	Metric	Short-scan FDK	PAR	2D+time U-Net	ConvLSTM NN	VRT	TT U-Net (proposed)
Case 1	RMSE-Local(HU)↓	57.96±30.26	47.39±21.69	24.99±10.55	23.26±9.31	24.73±11.20	**21.10±7.33**
	SSIM-Local(%)↑	67.93±13.22	71.82±14.04	81.84±8.69	83.02±7.81	81.84±8.53	**84.52±7.04**
	Dice-RCA(%)↑	58.22±29.73	65.23±22.87	76.66±7.47	76.86±7.28	79.65±4.11	**81.30±2.30**
Case 2	RMSE-Local(HU)↓	51.49±21.85	38.33±15.60	21.54±10.32	20.44±8.77	19.64±8.32	**19.24±7.97**
	SSIM-Local(%)↑	67.85±11.71	73.26±11.51	80.95±8.29	81.97±7.55	81.25±8.52	**83.16±6.94**
	Dice-RCA(%)↑	32.16±22.84	49.95±16.35	55.79±13.88	57.86±10.00	**62.00±5.16**	61.84±5.30
Case 3	RMSE-Local(HU)↓	59.93±27.52	51.31±23.74	26.26±11.26	24.03±9.16	23.16±8.89	**21.87±7.74**
	SSIM-Local(%)↑	67.26±11.46	71.83±11.88	83.36±6.91	84.48±6.25	82.59±7.64	**85.85±5.88**
	Dice-RCA(%)↑	61.08±22.15	71.92±10.41	77.40±7.58	77.80±6.28	80.23±3.62	**82.22±2.68**

the red arrow is distorted and some of the complex structures are blurred. Although the XCAT phantom provides dynamic images, the huge distribution gap between XCAT phantoms and clinical images may lead to the problem of domain shift. In contrast, the proposed PAD with a realistic appearance will be preferable for deep learning research.

An extra clinical case is presented in Fig. 8 where multiple calcified lesions are seen in the left anterior descending artery. The typical crescent motion artifacts are visible in the ROI image, and the dense calcified lesions show higher CT values. Our TT U-Net can remove motion artifacts while keeping the shape and CT value of the lesion in a post-processing way, even though coronary calcifications have not been included in the PAD for now. It shows the potential for our model to deal with such unseen cases.

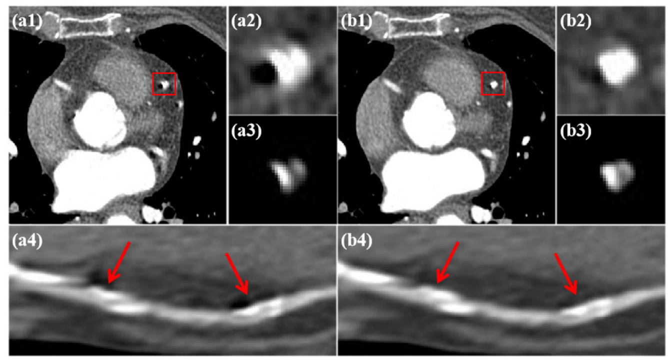

Fig. 8. Clinical case 5 with coronary calcifications. (a1)-(a3) The uncorrected image with zoom-in ROIs, (a4) Multiplanar reformation (MPR) of the uncorrected image, (b1)-(b3) TT U-Net corrected image with zoom-in ROIs, (b4) MPR of the corrected image. The display window is [-300,500] HU for reconstructed images, [-300,500] HU for (a2) and (b2) to visualize the motion artifacts, and [200,1000] HU for (a3) and (b3) to visualize the calcified lesion.

B. Quantitative Evaluation on the PAD

The methods were also quantitatively evaluated on three simulated cases from the PAD. We used the RMSE and SSIM metrics to evaluate the performance of myocardium motion artifact reduction, and the Dice score to evaluate the performance of RCA reconstruction. Table V shows the average quantitative metrics. As the intensity of cardiac motion varies from phase to phase, we plotted the metric curves for better visualization in Fig. 9. Generally, more significant improvements are seen at rapid cardiac phases. The PAR method is able to improve image quality at quasi-quiescent periods (approximately 40% and 75% of the cardiac cycle). However, it can hardly compensate for cardiac motion at rapid phases. On the other hand, all the deep learning methods achieve competitive results. It shows that deep neural networks trained on the PAD have great power in performing the deblurring task. However, the recovery of detailed structures requires extra temporal information, especially for fast-moving RCAs. In Fig. 9(c1), (c3), the Dice scores of the 2D+time U-Net and the ConvLSTM NN drop dramatically during the rapid phases. In contrast, our TT U-Net achieves significantly lower standard deviations of the Dice score, and the Dice score is almost consistent across the whole cardiac cycle for cases 1 and 3. This demonstrates the superior performance of the proposed network model.

C. Ablation Study

To evaluate the effectiveness of the proposed TTL module, we conducted three ablation studies and compared the performance of models with different network structures. First, we present the temporal profile results in Fig. 10 to intuitively show the impact of the TTL. The temporal profile helps to present the cardiac motion in a 2D image. Due to rapid cardiac motion, the temporal profile of the uncorrected image is blurry. Severe temporal inconsistency is also observed for the model without TTL. In contrast, the proposed TT U-Net provides much clearer results.

In the second experiment, we focus on the influence of the temporal window size (TWS). The TT U-Net computes self-attention within the TWS. Thus, by adjusting the TWS, we modulate the model's capacity for temporal modeling. The quantitative metrics are shown in Table VI. Note that we kept the 3D decoder for all the variants as used in the original TT U-Net to integrate temporal features. Compared with the w/o TTL" group, the performance of models with TTLs all improve by a large margin. This demonstrates that temporal information is important for motion artifact reduction. It also shows that the TT U-Net can adapt to a shorter acquisition time when the radiation dose becomes the major concern. On the other hand, we found that a longer sequence yields better performance, which suggests the benefit of direct interaction between distant frames. Because of limited computational resources, we set the maximum TWS to 48. We believe the TT U-Net can be adapted to a longer sequence if necessary.

In the third experiment, we investigate the integration strategy of the TTL module and the U-Net architecture.

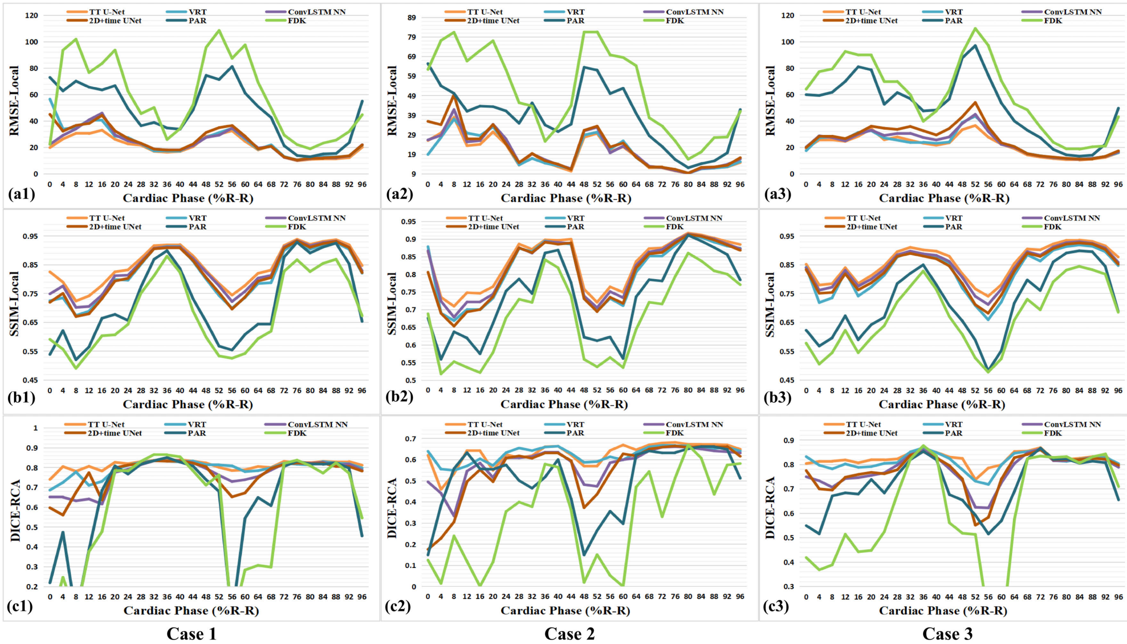

Fig. 9. Quantitative evaluations varying with cardiac phase for each simulated case. (a1)-(a3) The RMSE curves of various reconstruction methods, (b1)-(b3) the SSIM curves of various reconstruction methods, (c1)-(c3) the DICE curves of various reconstruction methods.

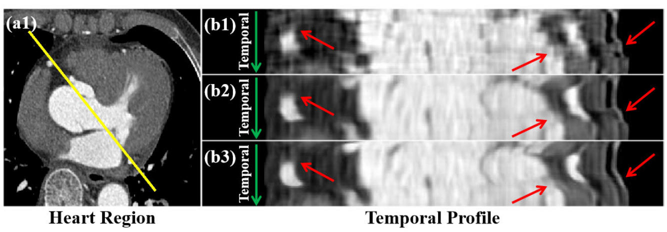

Fig. 10. Illustration of the impact of TTL. (a1) The heart region, (b1) the temporal profile of uncorrected clinical cardiac CT image, (b2) the correction result of the model without TTL, (b3) the correction result of the model with TTL. The yellow line denotes the spatial coordinates of the profile. The green arrow denotes the temporal dimension. Inconsistencies along the temporal dimension are denoted by red arrows.

In general, the encoder part of the TT U-Net is inherited from the original U-Net to extract spatial features at different spatial resolutions. The TTL modules thereafter encode the temporal features based on the hierarchical feature maps. The quantitative metrics in Table VII evaluate the effectiveness of each TTL. TTL1, TTL2, TTL3, and TTL4 correspond to the TTLs at different stages, where 1 refers to the lowest spatial resolution and 4 refers to the highest spatial resolution. TTL1 achieves substantial performance gain, whereas the performance saturates in the presence of TTL4. As the input of TTL1 comes from the deeper layers of the encoder path, it contains higher-level spatial features and has a larger receptive field to better detect streak motion artifacts. Additionally, the high-dimensional feature map exhibits a small spatial size, which enables us to scale our TT U-Net to a larger model. Thus, TTL1 largely improves the performance of the model. In contrast, the TTL4 has a relatively small spatial receptive field and contains fewer trainable parameters. To better meet the clinical demand for faster inference, we omit TTL4 in TT U-Net. However, the TTL can still be extended to even higher resolution and be effective in other tasks.

TABLE VI
AVERAGED RESULTS FROM THE ABLATION STUDY ON THE TEMPORAL WINDOW SIZE

Simulated Case	Metric	Temporal Window Size (TWS) w/o TTL	TWS = 12	TWS = 24	TWS = 48
Case 1	RMSE-Local(HU)↓	38.91	24.47	22.11	**21.10**
	SSIM-Local(%)↑	73.36	81.28	82.85	**84.52**
	Dice-RCA(%)↑	76.12	80.35	80.12	**81.30**
Case 2	RMSE-Local(HU)↓	38.45	22.07	20.73	**19.24**
	SSIM-Local(%)↑	71.10	80.48	81.54	**83.16**
	Dice-RCA(%)↑	52.13	60.89	60.87	**61.84**
Case 3	RMSE-Local(HU)↓	29.74	26.05	23.19	**21.87**
	SSIM-Local(%)↑	81.00	82.46	83.89	**85.85**
	Dice-RCA(%)↑	73.58	79.31	79.79	**82.22**

Note: The TWS = 48 group corresponds to the original TT U-Net.

V. DISCUSSIONS

A. Does TT U-Net "Read" Dynamic CT Images?

Previous research has demonstrated the relationship between the shapes of motion artifacts and the CT scan geometry [18]. One interesting finding is that CT systems rotating in reverse directions present either spatially or temporally flipped motion artifacts (see Fig. 11). It reminds us of the study of visual chirality [41]. In the context of dynamic CT imaging, motion artifacts are also chiral. As chirality appears on dynamic CT images, we call it "motion chirality". Based on this observation, we conducted an experiment to show that our TT U-Net is aware of motion.

Two TT U-Nets were trained on a normal PAD and a flipped PAD. Except for the CT rotation direction, all the other settings remained the same. We then evaluated the performance of the two trained models on a clinical dataset. The TT U-Net reduced most of the artifacts as expected in Fig. 12(b1)-(b4). However, severe performance degradation is observed with the model trained on the flipped dataset (see

TABLE VII
AVERAGED RESULTS FROM THE ABLATION STUDY ON TTLS AT DIFFERENT STAGES

Simulated Case	Metric	w/o TTL	TTL 1	TTL 1&2	TTL 1&2&3	TTL 1&2&3&4
Case 1	RMSE-Local(HU)↓	38.91	23.30	21.25	21.10	**20.91**
	SSIM-Local(%)↑	73.36	82.86	83.58	84.52	**84.62**
	Dice-RCA(%)↑	76.12	80.35	81.21	**81.30**	81.01
Case 2	RMSE-Local(HU)↓	38.45	20.45	19.69	19.24	**19.22**
	SSIM-Local(%)↑	71.10	82.14	82.56	83.16	**83.29**
	Dice-RCA(%)↑	52.13	60.95	62.02	61.84	**62.06**
Case 3	RMSE-Local(HU)↓	29.74	23.43	22.40	21.87	**21.85**
	SSIM-Local(%)↑	81.00	84.14	84.87	85.85	**86.00**
	Dice-RCA(%)↑	73.58	80.52	81.93	**82.22**	82.09

Note: TTL1, 2, 3, and 4 refer to the TTLs at different stages, where 1 is the lowest spatial resolution, and 4 is the highest spatial resolution.

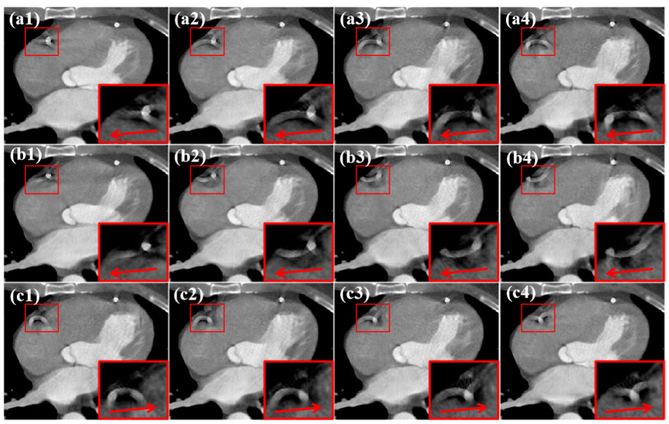

Fig. 11. Motion chirality. (a1)-(a4) Anti-clockwise scan with reference motion artifacts, (b1)-(b4) clockwise scan with spatially flipped motion artifacts, (c1)-(c4) clockwise scan of an RCA moving in a reverse direction, the motion artifacts are spatially similar to the reference artifacts but temporally flipped. The coronary motion direction is noted by red arrows.

Fig. 12(c1)-(c4)). In fact, although visually similar for motion artifacts on each single-phase image, the motion chirality leads to completely different spatiotemporal features for dynamic images. Therefore, the model trained on a flipped dataset does not generalize well on clinical images with reverse rotation direction. This phenomenon suggests that the TT U-Net estimates cardiac motion not only based on the artifact shape but also on the way it evolves over time. Namely, the trained model can take the spatiotemporal information into account and which helps reduce ambiguity. In this way, the model can better distinguish pathological changes in tissues (*e.g.*, coronary calcifications in Fig. 8) from extreme CT values induced by motion artifacts according to the different motion patterns.

B. Future Works

In this paper, we proposed a 2D+time TT U-Net for motion artifact reduction. In fact, the cardiac motion is in 4D (3D+time), and a 4D model is expected to exhibit higher performance. In our future work, we aim to further investigate the cardiac CT imaging problem directly in the 4D space. A larger dataset is required to train a 4D network model. As the proposed PAD is easily scalable, we believe it may

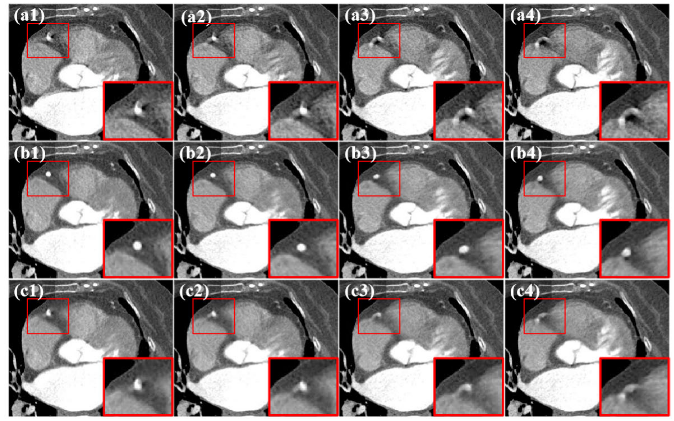

Fig. 12. Clinical case 3, performance evaluation of models trained on different datasets. (a1)-(a4) Uncorrected images, (b1)-(b4) corrected images using TT U-Net trained on a normal dataset, (c1)-(c4) corrected images using TT U-Net trained on a flipped dataset.

fulfill the need for a large dataset. It is also nontrivial to lift a low-dimensional network model to 4D because of the curse of dimensionality. The proposed TT U-Net is a possible candidate, as it processes the temporal and spatial features separably for efficient encoding. However, the high redundancy of the 4D cardiac CT image calls for an elaborate design of data representation and model optimization. We plan to design a model that can process 4D cardiac CT images more effectively and efficiently.

In addition, although we have shown that our method works well on real CT scans, cardiac CT imaging faces various difficulties in clinical practice. The high heart rate and irregular heart rhythm of patients may introduce complicated motion artifacts. Unexpected image degradations (*e.g.*, calcium blooming and the hybrid artifacts induced by a moving pacemaker) are also seen on cardiac CT images. In general, clinical cardiac CT images follow a long-tailed distribution, and efforts are needed to further adapt our method to various situations. As a preliminary study of deep learning based motion artifact reduction in cardiac CT, we built the PAD to enable a supervised learning framework. We plan to further improve the diversity of the PAD to train a more robust model.

Inspired by the recent study of stabilizing deep tomographic reconstruction [42], an alternative way to mitigate the above two problems is to incorporate our TT U-Net with the traditional MCR method. For example, the trained network model can serve as a deblurring module and enable better motion estimation. The deformation field estimated from deblurred images will be more accurate and thus benefit the MCR step. From another perspective, the MCR method introduces a correction loop for the data-driven model and enhances interpretability. By performing these two steps iteratively, we have seen encouraging results in our preliminary experiments.

VI. CONCLUSION

In this paper, we proposed a deep learning based framework for motion artifact reduction in dynamic cardiac CT. To train a deep neural network model for motion artifact reduction, we built a PAD consisting of paired reference and artifact-perturbed cardiac CT images. The effectiveness of the PAD was proven by experiments on clinical CT scans. For the model architecture, we designed a TTL module to extract both the local and global temporal features in dynamic CT images. With the temporal self-attention mechanism, the proposed TT U-Net was able to reduce the motion artifacts across the whole cardiac cycle in a post-processing way. Our framework is effective and extendable, and we believe it can benefit the broader research on dynamic medical imaging.

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